SinoGNSS

PRODUCT SPECIFICATION

For

K902 GNSS Module



INTRODUCTION

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REVISION HISTORY

Revision	Modification	Date
1.0	New Release	2024/09
1.1	Add an Identification of PIN 1	2024/12
1.2	Add Usage Requirements of Pin SYS_RSTN	2024/12
1.3	Add Transformer Connection Mode	2025/02
1.4	Modify the Protect Image	2025/03
1.5	Modify the pin definitions	2025/06
1.6	Optimize the document structure and modify some parameters	2025/06
1.7	Precautions for Increasing SMT Soldering Temperature	2025/12

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1. Introduction

K902 is a small-size high-precision positioning module independently developed by SinoGNSS at full - system and full-frequency. It tracks constellations including BDS-2, BDS-3, GPS, GLONASS, Galileo and QZSS, also with onboard inertial navigation device, supporting integrated navigation algorithm. The K902 GNSS module is mainly used for surveying and mapping, monitoring, outdoor robotics and so on.

The K902 module is based on the new-generation integrated high-precision GNSS navigation and positioning SoC chip from SinoGNSS. It adopts a 22nm low-power process, has a dual-core processor, 1688 channels, and is equipped with broadband signal reception technology, narrowband anti-interference technology, anti-continuous wave interference technology, multi-system joint positioning technology, ground-based/satellite-based augmentation technology, SBAS technology, and integrity technology. It possesses a more powerful satellite signal processing capability, providing users with highly reliable location information.

1.1. Product Characteristics

Table 1.Product Characteristics

Characteristics	Technical index	
Signals	Positioning	BDS-2: B1I, B2I, B3I BDS-3: B1I, B1C, B2a, B2b, B3I
		GPS: L1C/A, L2P, L2C, L5, L1C
		GLONASS: G1, G2, G3*
		Galileo: E1, E5a, E5b, E5 AltBoC, E6c
		QZSS: L1C/A, L1C, L2C, L5
		SBAS: L1C/A, L5
		NavIC: L5*
		L-Band*
	Items with * will be adjusted with the version.	
Time to First Fix	Cold Start	20s
	Hot Start (with RTC)	10s
Signal Capture	Reacquisition	1s
	Signals Tracking Sensitivity	-160dBm
	Signal Capture Sensitivity	-145dBm

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Measurement Precision	Pseudo-range Accuracy	10cm
	Carrier Phase Accuracy	1mm
Accuracy	PPS（RMS）	20ns
	SPP Accuracy	Horizontality：1.5m Verticality：3m
	Velocity Accuracy	0.02m/s
	Assisted dead reckoning (error)	3%d (d = driving distance)
PPP	PPP Convergence Time	20min
	PPP Accuracy	Horizontality：0.1m Verticality：0.2m
RTK	RTK Initialization Time	5s（D<10km）
	Initialization Reliability	99.9%
	RTK Accuracy	Horizontality：$(8+10^{-6}\times D)$mm Verticality：$(15+10^{-6}\times D)$mm D-Baseline length (Unit: mm)
Anti-interference	It can suppress the potential narrowband and single tone radio interference signals in the GNSS signal frequency band, and the interference to signal ratio can reach 60dB.	
IMU	Gyroscope	Measurement ranges：±125°/s
		Zero-biased repeatability：0.5°/s
		Zero-biased stability：5°/h
		Angular random walk：0.12°/sqrt(h)
	Accelerometer	Measurement ranges：±2g

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		Zero-biased repeatability：20mg
		Bias instability：50ug
		Velocity random walk：0.07m/s/sqrt(h)
Data Rates	Position	The default is 10Hz, and the maximum option is 50Hz
	RTK	The default is 10Hz, and the maximum option is 50Hz
	IMU	The default is 10Hz, and the maximum option is 50Hz
Data Formats	NMEA-0183	GPGGA, GPGSV, GPGLL, GPGSA, GPGST, GPHDT, GPRMC, GPVTG, GPZDA etc.
	ComNav Binary（CNB）	ComNav Self-Defined Binary
	CMR(GPS)	CMROBS, CMRREF
	RTCM2.X	RTCM1, RTCM3, RTCM9, RTCM1819, RTCM31, RTCM41, RTCM42
	RTCM3.X	1004 ~ 1008, 1012, 1019, 1020, 1033, 1042, 1045/1046, 1230, 4078 MSM3~MSM7: 1073~1077, 1083~1087, 1123~1127, 1093~1097
Electrical	Voltage	+3.1V~3.5V DC
	Power Consumption	0.41W (Anti-interference off) Set anti-interference on consumes more about 0.1W
Environmental	Operating Temperature	-40℃~+85℃ Attention: The module's heat dissipation pad must be effectively grounded, and the heat dissipation area must be no less than the area of the module.
	Storage Temperature	-55℃~+95℃
Hardware Interface		UART×3，CAN*×1（Option）

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		I^2C×1, SPI×1, PPS×1, EVENT×1
		* CAN and UART3 are multiplexed. By default, UART3 is used.
Physical	Size	17mm×22mm×2.5mm
	Weight	1.9g
	Package	LGA (54PIN)

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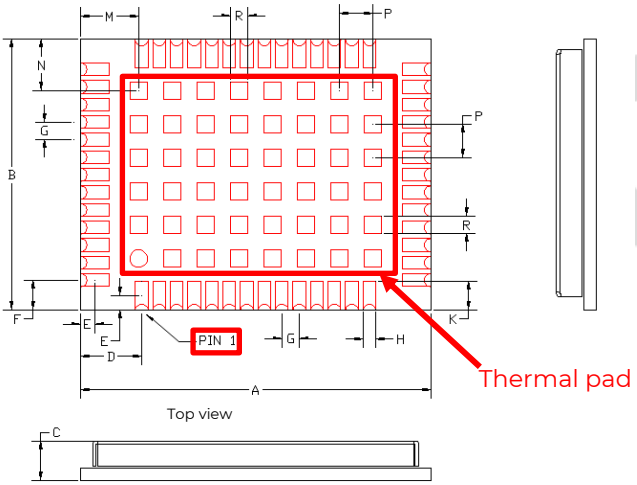
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2. Size

In this section, product photo, mechanical drawing and the dimension of K902 is provided for customers' further hardware design and installation.



Figure 1. K902 Product Photo



No.	Numerical value (mm)	Tolerance (mm)
A	22.00	±0.2
B	17.00	±0.2
C	2.50	±0.2
D	3.85	-
E	0.90	-
F	1.90	-
G	1.10	-
H	0.80	-
M	3.65	-
N	3.25	-
R	1.10	-
P	2.10	-
K	1.80	-

Figure 2. K902 Mechanical Drawing

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3. Pin Arrangement and Definition

K902 is surface-mount OEM Module which integrates 54 Pin.

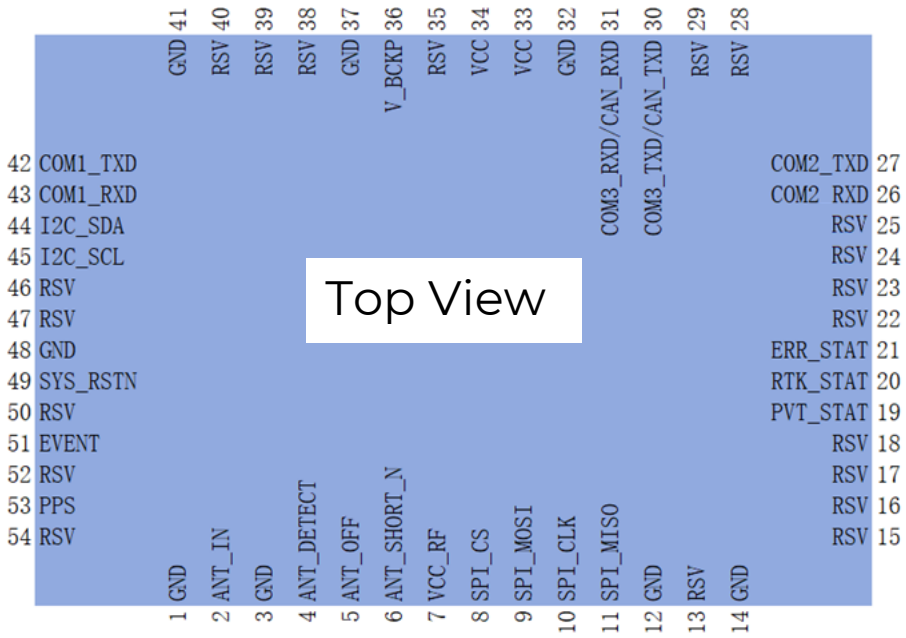


Figure 3. K902 Includes 54-Pin Pad

Table 2. Pin Definition of K902 54-Pin Pad

Pin	Signal	Type	Description
1	GND	GND	Ground Reference
2	ANT_IN	I	ANT input
3	GND	GND	Ground Reference
4	ANT_DETECT*1	I	Active antenna detect - default active high
5	ANT_OFF*1	O	External LNA disable - default active high
6	ANT_SHORT_N*1	I	Active antenna short detect - default active low
7	VCC_RF*2	O	Voltage for external LNA
8	SPI_CS	I/O	SPI_CS
9	SPI_MOSI	I/O	SPI_MOSI
10	SPI_CLK	I/O	SPI_CLK
11	SPI_MISO	I/O -	SPI_MISO
12	GND	GND	Ground Reference
13	RSV	-	Reserved
14	GND	GND	Ground Reference
15-18	RSV	-	Reserved
19	PVT_STAT	O	Number of search satellites
20	RTK_STAT	O	RTK Data Indicator
21	ERR_STAT	O	Failure Indication
22-25	RSV	-	Reserved
26	COM2_RXD	I	UART2 receive
27	COM2_TXD	O	UART2 transmit

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Pin	Signal	Type	Description
28-29	RSV	-	Reserved
30	COM3_TXD/CAN_TXD	O	UART3/CAN transmit
31	COM3_RXD/CAN_RXD	I	UART3/CAN receive
32	GND	GND	Ground Reference
33~34	VCC	PWR	Voltage supply
35	RSV	-	Reserved
36	V_BCKP	PWR	Backup supply voltage
37	GND	GND	Ground Reference
38-40	RSV	-	Reserved
41	GND	GND	Ground Reference
42	COM1_TXD	O	UART1 transmit
43	COM1_RXD	I	UART1 receive
44	I²C_SDA	I/O	Data Line
45	I²C_SCL	I/O	Clock Line
46-47	RSV	-	Reserved
48	GND	GND	Ground Reference
49	SYS_RSTN	I	System Reset Negative
50	RSV	-	Reserved
51	EVENT	I	External Interrupt Pin
52	RSV	-	Reserved
53	PPS	O	Pulse Per Second
54	RSV	-	Reserved

Note:
*1. It is not supported temporarily;
*2. It is not recommended to use VCC_RF as ANT_BIAS to feed power to the antenna (due to the limited module size, VCC_RF is not equipped with lightning protection, surge protection, and overcurrent protection).

3.1. Remarks

1. Electrical Characteristics

Table 3. Maximum withstand value

Specification	Symbol	Minimal value	Maximum value	Unit
Supply voltage (VCC)	VCC	-0.3	3.6	V
Input pin voltage	Vin	-0.3	VCC+0.3	V

Table 4.Operating Conditions

Specification	Symbol	Minimal value	Typical value	Maximum value	Unit	Condition
Supply voltage (VCC)	VCC	3.1	3.3	3.5	V	
Maximum VCC Ripple	V_{rpp}	0		50	mV	
Operating Current	I_{opr}		124		mA	VCC=3.3V
Power Consumption	P		410		mW	

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Table 5. I/O Threshold Characteristics

Specification	Symbol	Minimal value	Typical value	Maximum value	Unit	Condition
Input pin low level	V_{in_low}	-0.3		0.8	V	
Input pin high level	V_{in_high}	2		VCC+0.3	V	
Output pin low level	V_{out_low}	0		0.41	V	I_{out}=4mA
Output pin high level	V_{out_high}	VCC-0.4		VCC	V	I_{out}=4mA

Table 6. Antenna Characteristics

Specification	Symbol	Minimal value	Maximum value	Unit
Optimal input gain value	G_{ant}	20	35	dB

2. Power On and Off the module

VCC/V_BCKP

- VCC/V_BCKP power-on, the initial level is lower than 0.4V, and has a good monotonicity, and the upstroke and ringing are guaranteed within the range of 5% VCC/V_BCKP;
- Power-on interval，after the VCC/V_BCKP is powered off less than 0.4V, before the next power-on, the interval must be greater than 500ms.

3. Usage Requirements of Pin SYS_RSTN

- The module is equipped with an internal power-on reset circuit, eliminating the need for an external hardware reset circuit;
- If choose to control the reset of SYS_RSTN through GPIO, ensure that VCC has been properly powered up; then use GPIO to reset SYS_RSTN by pulling it down to ground (longer than 1ms) and then pulling it up to 3.3V.

4. RTK_STAT

RTK_STAT signal pin, receiving or transmitting differential signals will output high level pulse, other states output low level, needing to connect to the external indicator light.

5. Hardware Integration Considerations

1) Use the VCC pin to provide a reliable power supply and all GND pins of the module are grounded;
2) Connect the ANT_IN signal to the antenna, pay attention to the 50Ω impedance matching of the line;
3) The module reset pin SYS_RSTN is for reset. Please connect it correctly to ensure that the module can be reset reliably;
4) Special attention should be paid to:
   ①. Place the power supply as close to the module as possible
   ②. Widen the power supply lines or use split copper surface to transmit current;
   ③. Power routing should avoid high-power and high-inductive devices, such as magnetic coils.
   ④. To ensure good heat dissipation of the module, it is recommended that the grounding pad of the module be fully in contact with the PCB and that more ground holes be added to facilitate heat dissipation
5) Antenna interface: The antenna line is as short and smooth as possible, avoiding acute angles;
6) Avoid wiring directly under K902;
7) The module is as far away from the high temperature airflow as possible.

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8) When the user integrates the module, an external high-gain choke antenna is connected. If a surge protector is not added, it is easy to damage antenna feed circuit. It is recommended that users install a surge protector on the antenna when connecting a high-gain choke antenna;
9) The K902 module contains temperature-sensitive components inside, requiring a stable temperature environment; hence, during PCB layout, the module needs to be positioned away from heat-generating or low-temperature areas. The TCXO (Temperature Compensated X Tal Oscillator) inside the module is highly sensitive to temperature changes, and sudden temperature fluctuations can affect the module's ability to track satellite signals. Consequently, the layout of the module should maintain a certain distance from heat-generating components, cooling fans, or heat dissipation vents.

6. Static Electricity Protection

Some components on K902 module are easily damaged by static electricity, which in turn affects the IC circuit and other components. Therefore, you should pay attention to electrostatic protection measures when using it.

1) When handling the module, try to wear gloves or finger cots and an anti-static wrist strap that meets the electrostatic protection standards;
2) During the process of taking the module, only the edge of the board should be taken, and the solder joints, circuit parts or components should not be directly touched to avoid sweat fingerprints from contaminating the solder joints;
3) The module should be protected by a soft protective pad between the module and the module interval during transportation;
4) When the module is idle, it should be placed on a soft protective pad (such as anti-static sponge pad, etc.), and do not stack at will;
5) The modules should be placed neatly and orderly, with a certain interval between modules to avoid collision with each other;
6) The module should be handled with care during use to prevent the module from being damaged by rough operation;
7) When powering on, pay attention to the positive and negative poles of the power supply and the voltage to avoid reverse connection and excessive voltage from burning the module;
8) When soldering the module to the motherboard, please ensure that the GND is soldered first, and then the ANT_IN pin;
9) When handling ANT_IN pin, do not touch any live capacitance or material (e.g., surface mount antenna, coaxial cable, wire iron, etc.) to avoid damage to the ANT_IN pin by the charge generated or stored by said capacitor or material;
10) Please make sure to solder the ANT_IN pin with an electrical protection iron.

4. Assembling & Repairing Note

4.1. Module Assembling Note

K902 is surface mounted, SMT welding is used for assembly.

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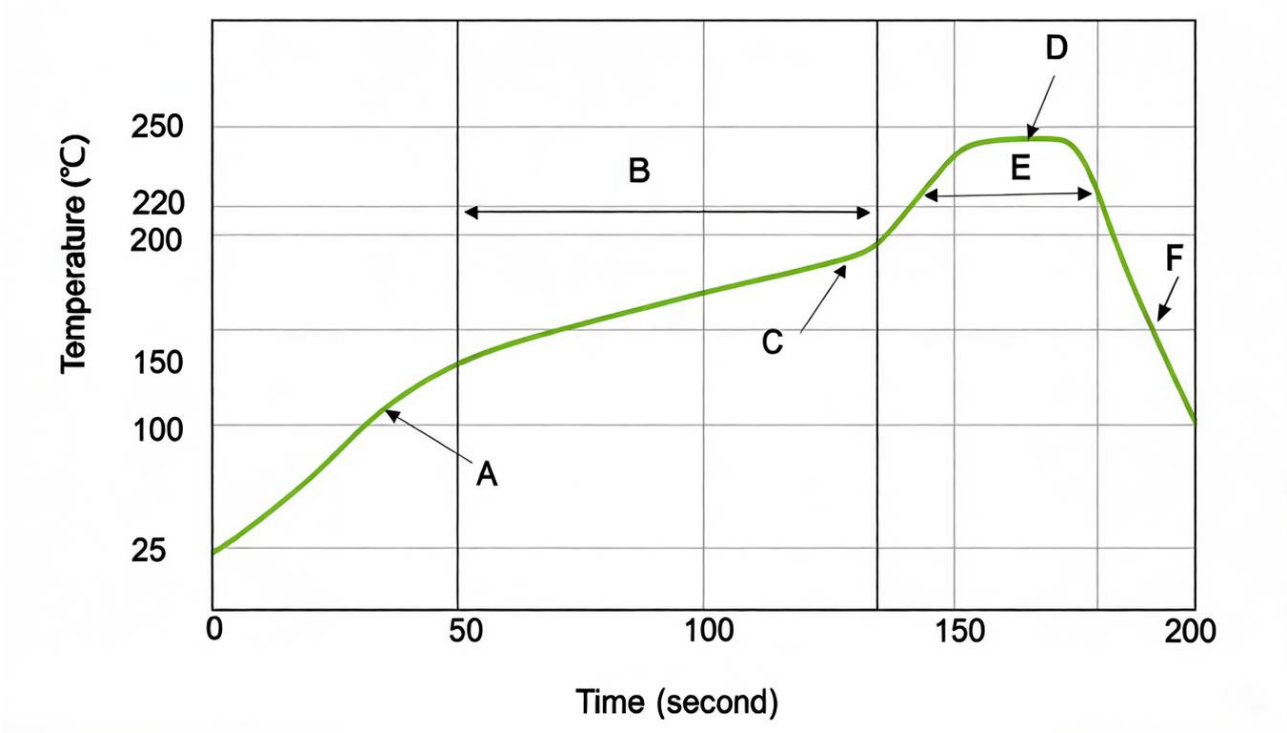


Figure 4. Furnace Temperature Curve

The process temperature limits are as follows:

A: Heating Zone: Rising Slope: 1 ~ 3℃ / sec

B: Constant Temperature Zone: Range: 150 ~ 190 ℃ , Time: 80 ~ 110 S

C: Constant Temperature→Reflow Zone: Rising Slope: 1 ~ 3℃ / sec

D: Peak Temperature: 235 ~ 245℃

E: Reflow Zone: Range: Over 220℃ Time: 50 ~ 80 S

F: Descent Slope: -5℃ ~ -1℃ / sec

Note: To prevent module damage caused by repeated heating, it is recommended to mount the module only after the reflow soldering of the first side of the PCB is completed. SMT soldering must be performed in strict accordance with the above temperature range. The peak temperature must not be exceeded, and a heat gun shall not be used for soldering. Otherwise, any resulting losses shall be borne by the user.

4.2. Repairing Note

When disassembling the module, it is suggested using a BGA welding bench. Please use correct air tuyere and choose certain temperature curve. Keep peak temperature under 245℃, rising slope under 3℃/s.

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5. Application Connection Example

5.1. Minimum Circuit Description

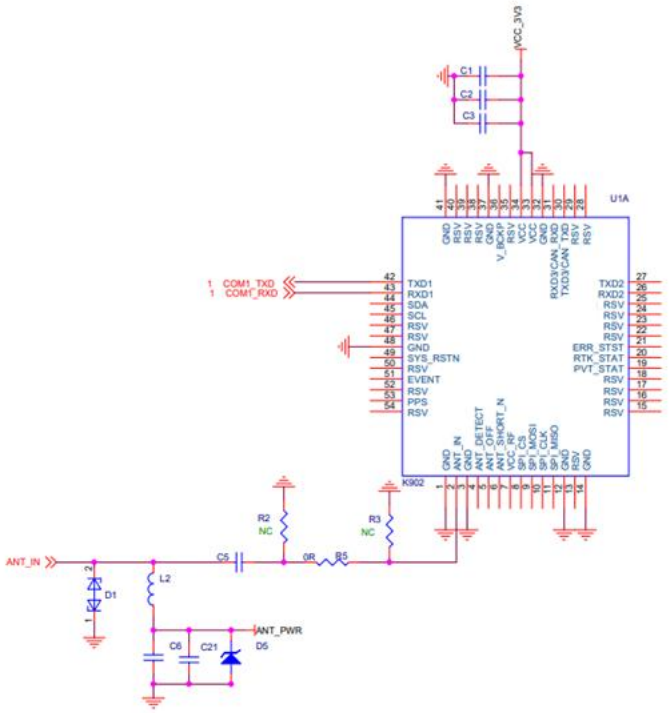


Figure 5. Minimum Hardware Design of K902

Table 7. Minimum Circuit Description of K902

Designator	Type	Value	Function
C1,C2	Decoupling Capacitor	22uF	Suppress low-frequency ripples and stabilize voltage.
C3	Decoupling Capacitor	1uF	Suppress high-frequency ripples and stabilize voltage.
C21	Decoupling Capacitor	0.1uF	Suppress high-frequency ripples and stabilize voltage.
C6	Decoupling Capacitor	1uF	Suppress low-frequency ripples and stabilize voltage.
L2	Feeding Inductor	68nH	RF Blocking, a 68nH RF inductor is recommended for use.
D1	ESD Diode	-	ESD protection device for anti-static purposes and support of high-frequency signals (above 2000MHz).
D5	TVS Diode	-	To prevent component damage caused by overvoltage, select a TVS diode with qualified clamping characteristics in accordance with the feed voltage, antenna voltage withstand and other specifications.
C5	DC Blocking	100pF	DC Blocking, Prevents Voltage Backflow Module.

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	Capacitor		
R2,R3	Resistor	-	A π-type circuit is reserved to ensure 50Ω impedance matching.
R5	Resistor	0Ω	Reserve component positions to ensure matching.

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6. Package

The K902 module is supplied in tape-and-reel packaging (compatible with mainstream surface mount equipment) and sealed in a vacuum antistatic aluminum foil bag (the shelf life of the K902 module in this antistatic bag is one year) with a desiccant included for moistureproofing. Moistureproof measures must still be implemented during transportation or when the module is unused after unpacking to prevent module failure. When soldering the module using the reflow soldering process, strictly comply with the IPC/JEDEC J-STD-033 standard for moisture control of the module. Since packaging materials such as the tape can only withstand a temperature of 55°C, the module must be removed from the packaging before baking.

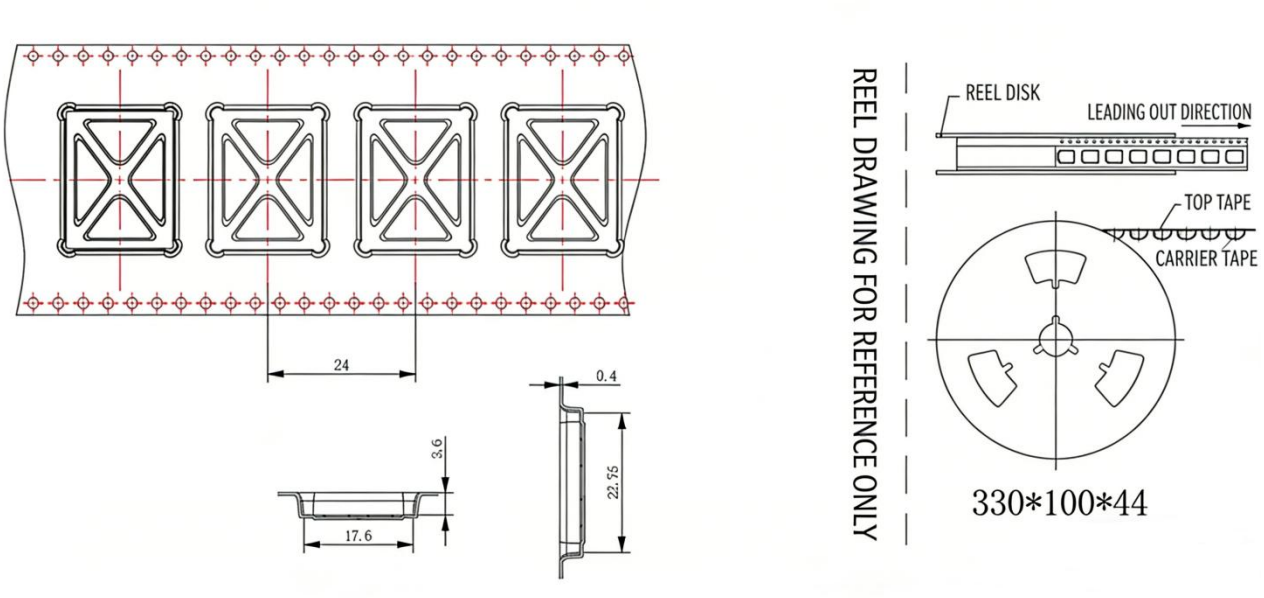


Figure 6. K902 Roll Tape Packing

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Table 8. K902 Package Description

Project		Description
Package	Number of Modules	500 Slice/Roll
	Reel Size	Material tray: 13 inches
		Outer diameter: 330mm, inner diameter: 100mm, width: 44mm, wall thickness: 3.6mm
		Package of each module: length 22.75mm, width 17.6mm
	Carrier Belt	Module Spacing (Center Distance): 24mm
	Quantity of Desiccant	1 pcs/box
	Number of Humidity card	1 piece/box Before surface mounting, check the humidity indicator card inside the package: the 30% indicator circle on the humidity card should normally appear blue. If the 20% indicator circle turns pink and the 30% indicator circle turns lavender, the module must be baked as required before mounting. The K902 module has a humidity sensitivity level (HSL) of 3. Packaging and handling precautions related to the humidity sensitivity level shall be in accordance with the IPC/JEDEC J-STD-033 standard.

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